

ДЗ 8

Вар 21

№	A	B
21.	4,7	0,2

Задание 1

Формат Ф1

$$A = (4.7)_{10} = (4.B(3))_{16} = (0.4B(3))_{16} * 16^1$$

$$B = (0.2)_{10} = (0.(3))_{16} * 16^0$$

$$X_C = X_A - X_B + d$$

$$X_C = 1 - 0 + 64 = 65$$

$$P_C = 1$$

№ шага	Действие	Делимое	Частное
0	M_A	0 0 1 0 0 1 0 1 1	0 0 0 0 0 0 0 0
	$[-M_B]_{\text{доп}}$	1 1 1 0 0 1 1 0 1	
	R_0	0 0 0 0 1 1 0 0 0	$R_0 > 0$
	$M_A \rightarrow 4$	0 0 0 0 0 0 1 0 0	1 0 1 1 0 0 0 0
	$[-M_B]_{\text{доп}}$	1 1 1 0 0 1 1 0 1	
	R_0	1 1 1 0 1 0 0 0 1	1 0 1 1 0 0 0 0
1	$\leftarrow R_0$	1 1 0 1 0 0 0 1 1	0 1 1 0 0 0 0 0
	$M_B \text{ пр}$	0 0 0 1 1 0 0 1 1	
	R_I	1 1 1 0 1 0 1 1 0	0 1 1 0 0 0 0 0
	$\leftarrow R_I$	1 1 0 1 0 1 1 0 0	1 1 0 0 0 0 0 0

2	M_B пр	0 0 0 1 1 0 0 1 1	
	R_2	1 1 1 0 1 1 1 1 1 _____↑	1 1 0 0 0 0 0 0 0
3	$\leftarrow R_2$	1 1 0 1 1 1 1 1 1	1 0 0 0 0 0 0 0 0
	M_B пр	0 0 0 1 1 0 0 1 1	
	R_3	1 1 1 1 1 0 0 1 0 _____↑	1 0 0 0 0 0 0 0 0
4	$\leftarrow R_3$	1 1 1 1 0 0 1 0 1	0 0 0 0 0 0 0 0 0
	M_B пр	0 0 0 1 1 0 0 1 1	
	R_4	0 0 0 0 1 1 0 0 0 _____↑	0 0 0 0 0 0 0 0 1
5	$\leftarrow R_4$	0 0 0 1 1 0 0 0 0	0 0 0 0 0 0 0 1 0
	$[-M_B]_{\text{доп}}$	1 1 1 0 0 1 1 0 1	
	R_5	1 1 1 1 1 1 1 0 1 _____↑	0 0 0 0 0 0 1 0 0
6	$\leftarrow R_5$	1 1 1 1 1 1 0 1 0	0 0 0 0 0 1 0 0 0
	M_B пр	0 0 0 1 1 0 0 1 1	
	R_6	0 0 0 1 0 1 1 0 1 _____↑	0 0 0 0 0 1 0 1 0
7	$\leftarrow R_6$	0 0 1 0 1 1 0 1 0	0 0 0 0 1 0 1 0 0
	$[-M_B]_{\text{доп}}$	1 1 1 0 0 1 1 0 1	
	R_7	0 0 0 1 0 0 1 1 1 _____↑	0 0 0 0 1 0 1 1 0
8	$\leftarrow R_7$	0 0 1 0 0 1 1 1 0	0 0 0 1 0 1 1 0 0
	$[-M_B]_{\text{доп}}$	1 1 1 0 0 1 1 0 1	
	R_8	0 0 0 0 1 1 0 1 1 _____↑	0 0 0 1 0 1 1 1 0

$$C^* = (1,7)_{16} * 16^1 = (17)_{16} = 23$$

$$\Delta C = C_T - C^* = 23.5 - 23 = 0.5,$$

$$\delta C = \left| \frac{\Delta C}{C_T} \right| \cdot 100\% = \left| \frac{0.5}{23.5} \right| \cdot 100\% = 2.13\%.$$

Погрешность вызвана неточным представлением операндов.

Формат Ф2

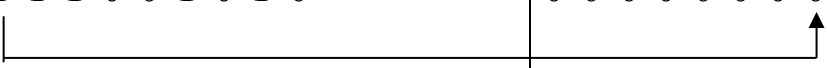
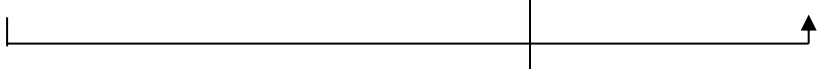
$$A = (4.7)_{10} = (0.100101100)_2 * 2^3$$

$$B = (0.2)_{10} = (0.1100110011)_2 * 2^{-2}$$

$$X_C = X_A - X_B + d$$

$$X_C = 3 + 2 + 128 = 133$$

$$P_C = 5$$

№ шага	Действие	Делимое	Частное
0	M_A	0 1 0 0 1 0 1 1 0	0 0 0 0 0 0 0 0
	$[-M_B]_{\text{доп}}$	1 0 0 1 1 0 1 0 0	
	R_0	1 1 1 0 0 1 0 1 0	0 0 0 0 0 0 0 0
			
1	$\leftarrow R_0$	1 1 0 0 1 0 1 0 0	0 0 0 0 0 0 0 0
	M_B пр	0 1 1 0 0 1 1 0 0	
	R_1	0 0 1 1 0 0 0 0 0	0 0 0 0 0 0 0 1
			
2	$\leftarrow R_1$	0 1 1 0 0 0 0 0 0	0 0 0 0 0 0 1 0
	$[-M_B]_{\text{доп}}$	1 0 0 1 1 0 1 0 0	
	R_2	1 1 1 1 1 0 1 0 0	0 0 0 0 0 0 1 0
			
	$\leftarrow R_2$	1 1 1 1 0 1 0 0 0	0 0 0 0 0 1 0 0

3	M_B пр	0 1 1 0 0 1 1 0 0	
	R_3	0 1 0 1 1 0 1 0 0 _____	0 0 0 0 0 1 0 1 ↑
4	$\leftarrow R_3$	1 0 1 1 0 1 0 0 0	0 0 0 0 1 0 1 0
	$[-M_B]_{\text{доп}}$	1 0 0 1 1 0 1 0 0	
	R_4	0 1 0 0 1 1 1 0 0 _____	0 0 0 0 1 0 1 1 ↑
5	$\leftarrow R_4$	1 0 0 1 1 1 0 0 0	0 0 0 1 0 1 1 0
	$[-M_B]_{\text{доп}}$	1 0 0 1 1 0 1 0 0	
	R_5	0 0 1 1 0 1 1 0 0 _____	0 0 0 1 0 1 1 1 ↑
6	$\leftarrow R_5$	0 1 1 0 1 1 0 0 0	0 0 1 0 1 1 1 0
	$[-M_B]_{\text{доп}}$	1 0 0 1 1 0 1 0 0	
	R_6	0 0 0 0 0 1 1 0 0 _____	0 0 1 0 1 1 1 1 ↑
7	$\leftarrow R_6$	0 0 0 0 1 1 0 0 0	0 1 0 1 1 1 1 0
	$[-M_B]_{\text{доп}}$	1 0 0 1 1 0 1 0 0	
	R_7	1 0 1 0 0 1 1 0 0 _____	0 1 0 1 1 1 1 0 ↑

$$C^* = (0.1011110)_2 \cdot 2^5 = (10111.10)_2 = 23.5.$$

$$\Delta C = C_T - C^* = 23.5 - 23.5 = 0,$$

$$\delta C = \left| \frac{\Delta C}{C_T} \right| \cdot 100\% = \left| \frac{0}{23.5} \right| \cdot 100\% = 0\%.$$

Погрешности нет.

В формате Ф2 операнды представлены точнее и погрешность меньше.